4

The social structure of psychological experimentation

How the Leipzig research community organized itself

In the last section of chapter 2 we took note of the fact that the establishment of psychological laboratories involved the institutionalization of certain social arrangements. Psychological experimentation became a collaborative effort dependent on a division of labor among individuals who carried out different functions in the experimental situation. More specifically, a broad distinction emerged between those who acted as the human source of psychological data, the experimental subjects, and those who manipulated the experimental conditions, the actual experimenters. Although this distinction was essentially a response to the practical exigencies of late nineteenth-century "brass instruments" experimentation and not the product of deep reflection, it rapidly became traditionalized. To carry out sophisticated experiments with relatively complex apparatus, one now had to assign the function of data source and the function of experimental manipulation to different people.

What was not noticed at the time, or indeed for almost a century afterward, was that this arrangement created a special kind of social system – one of psychological experimentation. The interaction between subjects and experimenters was regulated by a system of social constraints that set strict limits to what passed between them. Their communication in the experimental situation was governed by the roles they had assumed and was hedged around by taken-for-granted prescriptions and proscriptions.

However, the specific features of this social system were not necessarily fixed. The basic division of labor between experimenters and subjects still left much room for local variation. For instance, there was nothing in the practical requirements of psychological experimentation that dictated a *permanent* separation of experimenter and subject roles. The practical exigencies that had led to a separation of these roles did not preclude the possibility that the same person might take one or the other role on different occasions. All that experimental practice entailed was the convenience of separating these roles for a particular experimental session; it did not necessarily decree that people could not exchange these roles from session to session. However, other factors might enter the picture to convert the occasional separation of experimenter and subject roles into a permanent separation. Whether this in fact happened is a matter for historical investigation.

What kind of factor might be expected to have an effect on the permanence of the division of labor in experimental situations? One such factor is quite obvious. Psychological experiments are not conducted in a social vacuum.[1] The participants in such experiments are not social blanks but enter the experimental situation with an already established social identity. It is of course impossible to insulate the social situation of the experiment from the rest of social life to the point where the participants' social status outside the experiment has absolutely no bearing on their interaction within the experimental situation. Who gets to play experimenter and who subject may have something to do with the roles that individuals play outside the experimental situation. Similarly, the possibility of exchanging experimental roles may depend on the social identity of those who are occupying those roles.

Apart from the likely influence of social factors that are clearly external to the experimental situation, there is the question of how the distribution of roles in the face-to-face laboratory situation ties in with other activities that form a necessary part of the social practice of investigation. The direct interaction of experimenters and subjects in the laboratory is only one part of the research process. Activity in the laboratory is embedded in a penumbra of other activities, like theoretical conceptualization, data analysis, and writing a research report. The fact that in the contemporary psychology experiment these activities are assumed to form part of the experimenter role should not lead us to presuppose that this was always the case. The domain of what necessarily belonged to the experimenter role and what to the subject role has in fact been subject to historical variation. Let us therefore return to Wundt's Leipzig laboratory to see how subject and experimenter roles related to various other functions entailed by the social practice of psychological investigation.

When we peruse the research reports published in Wundt's house organ, the *Philosophische Studien,* we find that the person who had authored the published account of the experiment had not necessarily functioned as the experimenter. For instance, Mehner published a paper on an experiment in which he had functioned solely as the only subject while two other persons had functioned as experimenters at various times.[2] Moreover, functioning as a source of psychological data was considered incompatible with functioning as a source of theoretical conceptualization. During the first years of his laboratory, Wundt appeared regularly as a subject or data source in the experiments published by his students, although he also contributed much of the theory underlying these experiments.[3] Interestingly, Wundt does not seem to have taken the role of experimenter. This suggests that the role of psychological data source was considered to require more psychological sophistication than the role of experimenter. The role of experimental subject was quite compatible with Wundt's status as head of the laboratory, but the role of experimenter could be left to his students.

If the role of functioning as source of psychological data was quite compatible with the functions of a scientific investigator, there was clearly no reason to make a permanent distinction between experimenter and subject roles in the laboratory situation. Indeed, it was not uncommon for Wundt's students to alternate with one another as stimulus administrators and as sources of data, sometimes in the same experiment.[4] The roles of subject and experimenter were not rigidly segregated, so that the same person could occupy both at different times. Their differentiation was regarded as a matter of practical convenience, and most participants in the laboratory situation could play either or both roles equally well.

The participants in these experiments clearly saw themselves as engaged in a common enterprise, in which all the participants were regarded as collaborators, including the person who happened to be functioning as the experimental subject at any particular time. Characteristically, authors of experimental articles would sometimes refer to the experimental subjects as co-workers (*Mitarbeiter*), especially when giving their names as part of the published account of the investigation.[5] Moreover, functioning as subjects and experimenters for one another would usually be part of a relationship that extended beyond the experimental situation. Those who participated in these situations generally knew each other as fellow students, as friends or friends of friends, or as professor and student. The experimental situation was not based on the interaction of strangers.

What we find in Wundt's Leipzig laboratory is a distinctly collaborative style of interaction among the participants in the experimental situation, not a situation characterized by fixed role and status differentials. (Of course, Wundt himself occupied a special position, but once he entered the experimental situation as a subject his responses were tabulated along with everyone else's and were not given any special importance.) In these early experiments the interaction of investigators with their human data sources has not been clearly separated from the interaction among the investigators. Most of the individuals who functioned as subjects in these experiments were members of the research community. Not all the subjects who appear in the research reports of the Leipzig group were students of psychology, but those who were not seem to have had a definite interest in the subject and generally some experience of psychological experimentation. So instead of a two-tiered social structure in which there is a clear distinction between one social interchange with experimental subjects and another with colleagues, there is a less differentiated situation in which, to a large extent, the scientific community is its own data source.

This style of psychological experimentation appears to have resulted from the convergence of a particular tradition of scholarly work with a special kind of research goal. The scholarly tradition was that associated with the German nineteenth-century university system. It was based on linking the training of an intellectual elite with the systematic production of new knowledge in the context of a collaborative research enterprise.[6] The research goal involved the analysis of generalized processes characteristic of the normal, mature human mind. Psychological experimentation was designed to investigate this object much as physiological experimentation was designed to analyze biological processes involved in the functioning of normal, mature organisms.

But just as the practice of introspection had helped to construct the object it was meant to investigate, so the new practice of psychological experimentation constructed its own object. Experimental subjects were not studied as individual persons but as examples that displayed certain common human characteristics. That is why the role of subject could be assumed by any member of the research community. They did not represent themselves but their common mental processes. These "elementary" mental processes, as Wundt called them, were assumed to be natural objects that could be studied independently of the whole personality. All that was necessary were the restricted conditions of the laboratory and a certain preparation of the subject. That preparation involved a specific kind of background and a suitably collaborative attitude. But this bringing together of restricted conditions of stimulation and response – of psychologically sophisticated background and of a readiness to play the subject role – amounted to a careful construction of the very object that the research program was designed to investigate. This did not make it any less "real," but it did mean that it would have to compete for attention with other psychological objects constructed in different ways.

Alternative models of psychological investigation

The model of experimentation inaugurated at Leipzig had a special significance for the early development of psychology as an academic discipline with scientific pretensions, but it was not the only model of psychological experimentation available at the time. The *clinical experiment* had goals and social arrangements of quite a different order. The first major research program that depended on this type of experimentation involved the experimental study of hypnotic phenomena. At exactly the same time that Wundt's Leipzig laboratory was getting under way, a group of French investigators embarked on the systematic use of experimental hypnosis as a tool of psychological research.[7] In their studies various psychological functions were investigated under conditions of experimentally induced hypnosis. Now, contrary to the practice in Leipzig, there was no interchange of experimental roles among the participants in these French studies but a clear and permanent distinction between experimenters and individuals experimented on. Experimenters remained experimenters and hypnotic subjects remained hypnotic subjects. Moreover, there was a glaring status difference between these male scientists and their generally female lay subjects. The distribution of social functions approximated the pattern of the typical modern experiment, with the function of providing the data source being strictly segregated in an experimental subject role.[8]

By 1890, Binet had turned from experiments on hypnotized subjects to experiments on infants.[9] This was possible without altering the essential social structure of the experimental situation. For the Wundtian experiment, on the other hand, this kind of extension of scope was not possible.

The clinical experiment had emerged in a medical context. Those who functioned as subjects in these experiments were identified by labels such as "hysterics" or "somnambulists." Where normal or "healthy" subjects were used, it was for purposes of comparison with diagnosed clinical cases who were the essential target of the research. The experimenters were individuals with a medical background. Before experimental sessions began, the experimenter and the subject were already linked in a physician–patient relationship, and the essential features of this relationship were simply continued into the experimental situation. The whole situation was defined in medical terms. A crucial feature of this definition was the understanding that the psychological states and phenomena under study were something that the subject or patient underwent or suffered.

In the context of the clinical experiment, we find the first consistent usage of the term "subject" in experimental psychology. These medically oriented experimenters quite spontaneously refer to the patients they experimented on as "subjects" (*sujet*), because that term had long been in use to designate a living being that was the object of medical care or naturalistic observation. This usage goes back at least to Buffon in the eighteenth century. Before that a "subject" was a corpse used for purposes of anatomical dissection, and by the early nineteenth century one spoke of patients as being good or bad subjects for surgery.[10] When hypnosis came to be seen as an essentially medical matter, which was certainly the case in the Paris of the 1880s, there was nothing more natural than to extend an already established linguistic usage to yet another object of medical scrutiny. However, within the medical context we immediately get the formulation "healthy subject" (*sujet sains*)[11] when it is a matter of comparing the performances of normal and abnormal individuals. From this it is a very short step to the generalized use of the term subject to refer to any individual under psychological investigation, a step quickly taken by Binet and others.

The English term subject had acquired similar medical connotations as its French equivalent. In the eighteenth century it was used to refer to a corpse employed for anatomical dissection and by the middle of the nineteenth century it could also mean "a person who presents himself for or undergoes medical or surgical treatment" – hence its use in the context of hypnosis.[12] In the English-language psychological literature, the first uses of the term subject occur in the context of experiments involving the hypnotic state.[13] It is interesting to note that at this time Stanley Hall uses the term subject in the context of his work in the area of experimental hypnosis but switches to the terms "percipient" and "observer" when reporting on his experimental work with normal individuals.[14]

J. McKeen Cattell appears to have been the first to use the English term subject in describing a psychological experiment involving a normal adult human data source. However, he is by no means sure of his ground, for in an 1889 paper coauthored by him we find the formulation "an observer or subject," with "subject" in inverted commas.[15] Putting the term subject in inverted commas, when used in this context, was quite common in the English literature of this time, indicating that this usage was of recent origin, possibly influenced by French models.[16]

Thus the earliest years of experimental psychology were marked by the emergence of two very different models of the psychological experiment as a social situation. The Leipzig model involved a high degree of fluidity in the allocation of social functions in the experimental situation. While participants in this situation could assume both the role of experimenter and that of subject, the latter role was, if anything, more important than the former. In the clinical experiment, by contrast, experimenter and subject roles were rigidly segregated. The experimenter was clearly in charge, and only he was fully informed about the significance of the experiment. Experimenter and subject roles were not exchangeable. The objects of study that these two experimental situations were designed to display were clearly different. The clinical experiment was meant to display the effects of an abnormal condition for the benefit of the informed experimenter and those who were able to identify with him. On the other hand, the Leipzig experiment was meant to display universal processes that characterized all normal minds and whose significance was therefore open to all. In the one case the object of investigation presupposed the asymmetry of the experimenter–subject relation, but in the other case it did not.

Although these two models of psychological investigation constituted the types whose fundamental divergence holds the greatest theoretical interest, several other models had begun to emerge before the end of the nineteenth century. The earliest of these first appeared in England. In 1884, at the International Health Exhibition in London, Francis Galton set up a laboratory for "testing" the "mental faculties" of members of the public. Here we have an investigative situation whose social structure is clearly different from that of either the Leipzig or the Paris model. The individuals investigated were not medically stigmatized but presumed to be ordinary members of the public. Yet their role and that of the investigator were definitely not exchangeable, and between these roles there was a clear status difference. The investigator was supposed to have some kind of expert knowledge about the individual tested, and he was willing to share this knowledge upon payment of a fee. Galton charged every person who availed himself of the services of his laboratory the sum of threepence in return for which that individual received a card containing the results of the measurements that had been made on him or her. The relationship operated on a "fee for service" principle, but the service was clearly not thought of as being medical in character. Galton referred to the individuals who presented themselves for testing not as subjects but as "applicants."

There was apparently no lack of "applicants" for Galton's services, over 9,000 having been tested by the time the exhibition closed.[17] This seems to indicate that there was something not altogether unfamiliar about the kind of interaction that Galton's laboratory invited, and that the situation was sufficiently meaningful to a large number of people, not only to induce them to participate but also to pay money for the privilege. Competitive school examinations would certainly have provided a general social model for anthropometric testing. But beyond this it is possible that a more specific social model for Galton's anthropometric laboratory was provided by the practice of phrenology, which, though by then discredited, had been widely relied upon a generation earlier. Phrenologists had long offered individuals the service of informing them about their "mental faculties" on the basis of measurements performed upon them. Some of Galton's "applicants" may not have regarded his service as being so very different from that provided by a familiar model. Even Queen Victoria had consulted a phrenologist about her children, and as a young man Galton himself had gone to a phrenologist and taken his report very seriously.[18] I am not of course suggesting an explicit resolve to imitate the phrenologist's practice, but rather the operation of an analogous social practice as an implicit model that made Galton's innovative procedure immediately acceptable.

A comparison of the social structure of Galton's investigative practice with that of the Leipzig or the clinical experiment reveals some divergent features. The utilitarian and contractual elements of Galton's practice are particularly striking. Galton was offering to provide a service against payment, a service that was apparently considered of possible value to his subjects. What he contracted to provide his subjects with was information about their relative performance on specific tasks believed to reflect important abilities.[19] His subjects would have been interested in this information for the same reasons that they or their parents were interested in the information provided by phrenologists: In a society in which the social career of individuals depended on their marketable skills any "scientific" (i.e., believed to be objective and reliable) information pertaining to these skills was not only of possible instrumental value to the possessors of those skills but was also likely to be relevant to their self-image and their desire for self-improvement.[20]

From Galton's own point of view the major hoped-for return on this large-scale investment in "anthropometric measurement" was certainly not the money but a body of information that might ultimately be useful in connection with his eugenics program. The practical implementation of this social program of selective human breeding would have been facilitated by a good data base on human ability. Galton's interests in this research situation were just as practical as those of his subjects, the difference being that while they were interested in their plans for individual advancement he was interested in social planning and its rational foundation.

Three models of practice compared

The object of investigation that Galton and his subjects created through their structured interaction in the testing situation was very different from the objects constituted by either the Leipzig or the clinical experiment, although it had elements in common with both. What Galton's testing situation produced was essentially a set of individual *performances* that could be compared with each other. They had to be *individual* performances – collaborative performances were not countenanced in this situation. At least the performances were defined as individual, and the fact that they were the product of a collaboration between the anthropometrist and his subjects was not allowed to enter into the definition. The performances therefore defined characteristics of independent, socially isolated individuals and these characteristics were designated as "abilities." An ability was what a person could do on his own, and the object of interest was either the individual defined as an assembly of such abilities or the distribution of performance abilities in a population. The latter was Galton's primary object of interest, the former that of his subjects.

Locating the object of research within an isolated individual person was of course a basic common feature of all forms of psychological investigation; in fact, this constituted it as psychological, rather than some other kind of investigation. However, the line of inquiry inaugurated by Galton was the most uncompromising in this respect. As we saw in the previous chapter, Wundt was uncomfortable about the way in which the experimental method isolates the individual from the social–historical context, so that he limited the reach of this method to the "simpler" mental processes. The French clinical experimentalists were interested in the interindividual process of suggestion, which they mostly regarded as a fundamental condition for the appearance of the abnormal states they wanted to investigate. But Galton's anthropometry was quite radical in its conceptual severing of the links between an individual's performance and the social conditions of that performance. It accomplished this by defining individual performances as an expression of innate *biological* factors, thereby sealing them off from any possibility of social influence.

The potential practical appeal of Galton's anthropometric method was based on three features. First was the radical individualism, which we have just noted meant that it claimed to be dealing with stable and unalterable individual characteristics that owed nothing to social conditions. This provided the necessary ground for supposing that the performance of individuals in the test situation could be used as a guide to their performance in natural situations outside the laboratory.

Second, there was the partly implicit element of interindividual competition built into the method. Of interest was not just one individual's performance but the relative standing of that individual in comparison with others. For this type of investigative practice any one episode of experimenter–subject interaction only obtained its significance from the fact that it was part of a series of such interactions, each of them with a different subject. In the other forms of investigative practice single laboratory episodes were also often part of a series, but here the series was constituted by a variation of conditions with the same subject. In the Galtonian approach the series of experimental episodes was essentially defined by the differences among the subjects. Of course, duplication of an experiment with different subjects was known both in the Leipzig laboratory and in clinical experimentation. But this was merely a replication of the experimental episode for the purpose of checking on the reliability of the observations made. For the practice of anthropometry, on the other hand, the set of experimental episodes with different subjects formed a statistically linked series, and it was this series, rather than any of the individual episodes, that formed the essential unit in this form of investigative practice. Only in this way was it possible to generate the desired knowledge object: a set of performance norms against which individuals could be compared.

Whereas the Leipzig style of psychological experimentation only made use of the division of labor between experimental and subject roles, the Galtonian approach added to this the multiplication of subjects as an inherent and necessary component of its method. This was closely connected to a third feature of its practical appeal, namely, the statistical nature of the information it yielded. To be of practical value, the comparison of individual performances had to be unambiguous, so that rational individual or social policy decisions could be based on it. The way to achieve this was by assigning quantitative values to performances, thus allowing each individual performance to be precisely placed in relation to every other performance. Therefore, the anthropometric approach was designed to yield knowledge that was inherently statistical in nature. For the Galtonian investigator the individual subject was ultimately "a statistic." This made him a very different kind of knowledge object than the "case" of clinical experimentation or the exemplar of a generalized human mind, as in the Leipzig experiments.

These differences in the status of the knowledge object were linked to certain features of the interaction between investigators and their human subjects. What was of particular relevance here was the bearing of the prior relationship between experimenters and subjects on the social system of experimenter–subject interaction. In Galtonian practice investigators and their subjects met as strangers, contracting to collaborate for a brief period of experimentation or testing. In the clinical experiment investigators and subjects generally extended an existing pattern of physician–patient relationships into the experimental situation. In the Leipzig laboratory, as we have seen, the participants were fellow investigators whose common interests and friendships often extended across several experiments.

Thus, in each case there was a certain fit between the kind of knowledge object sought for and the nature of the experimental interaction that was used to produce it: The superficial interaction of strangers in a limited, contractually defined relationship was the perfect vehicle for the production of a knowledge object whose construction required a series of brief encounters with a succession of human subjects.[21] Similarly, the collaborative interaction of colleagues in exchangeable subject and experimenter roles was nicely suited to the construction of a knowledge object defined in terms of common basic processes in a generalized adult mind. Finally, the phenomena produced by clinical experimentation depended in large measure on a tacit agreement among the participants to extend the medical context into the experimental situation.

Each type of investigative situation constitutes a coherent pattern of theory and practice in which it is possible to distinguish three kinds of interdependent factors. First of all, there is the factor of *custom* which provides a set of shared taken-for-granted meanings and expectations without which the interaction of the participants in the research situation would not proceed along certain predictable paths. Much of this preunderstanding among the participants depends of course on general cultural meanings, but what is of more interest in the present context are the local variations in these patterns. We have distinguished several varieties of custom that played a role in the emergence of independent styles of psychological investigation in different places. These were the customs of the German nineteenth-century university research institute, of scientific medical investigation, of competitive school examinations, and possibly of phrenological assessment. Here were the sources of models of social interaction that could be readily adapted to the purposes of psychological research. The existence of these models meant that psychological research could get started without having to face the impossible task of inventing totally new social forms. However, once it did get successfully started the new practice led to new variants of the old forms and established its own traditions.

The actual practices of psychological investigation, though derived from certain customary patterns, were not identical with these patterns and therefore constituted a second distinguishable element in the emerging local styles. A third element was constituted by the diverging knowledge interests that have already been alluded to. The three original forms of psychological research practice were by no means interested in the same kind of knowledge. There was a world of difference between knowledge of elementary processes in the generalized human mind, knowledge of pathological states, and knowledge of individual performance comparisons. From the beginning, psychological investigators pursued several distinct knowledge goals and worked within investigative situations that were appropriate to those goals. However, it would be quite inaccurate to think of their adoption of appropriate investigative situations as a product of deliberate rational choice. The knowledge goals that these investigators pursued were as much rooted in certain traditions as were the practices that they adopted. Intellectual interests and the practices that realized those interests were both generally absorbed as a single cultural complex of theory and practice. Each such complex then led to the construction of the kind of knowledge object it had posited.

Modern psychology began with several different models of what a psychological investigation might look like, and the differences among these models went rather deep. Psychological investigation only existed in a number of different historical incarnations. Although in certain locations a particular model of investigation might achieve an overwhelming predominance for a considerable period, this should not mislead us into equating a particular form of experimentation with experimentation as such. Questions about the scope and limits of experimentation in psychology have usually presupposed a particular variant of experimental practice. However, it is historically more meaningful to compare the implications of different forms of practice.

Social psychological consequences

One set of questions that is suggested by the historical analysis of psychological experiments concerns the social psychology of such experiments. If there were different models of psychological investigation, with different patterns of social interaction among the participants, then each such model is likely to have entailed its own social psychological problems. In historical perspective it is inadequate to take one type of investigative situation as

Figure 4.1. Wilhelm Wundt and assistants at the Leipzig laboratory (photograph ca. 1910)

Figure 4.2. A medical demonstration of hypnosis and grand hysteria; from a painting by André Pierre Brouillet, *The School of Jean-Martin Charcot*

Figure 4.3. Galton's anthropometric laboratory in London

representing *the* psychological experiment and to generalize from the social psychological factors in this situation to all types of psychological experimentation.

One also needs to distinguish between the intended and the unintended social features of experimentation. This is where the term "artifact" can be seriously misleading.[22] All results of psychological experiments are social artifacts in the sense that they are produced by the participants in certain specially constructed social situations. But insofar as this production corresponds to the intentions and plans of those in charge of the experiments, it is not thought of as artifactual. That epithet is commonly reserved for the unavoidable *unintended* social features of experimentation – for example, the suggestive influence of the experimenter's hopes and expectations, or the inappropriate anxieties of the subject. However, just as the planned social features of experimentation vary considerably in different models of the experimental situation, so do the unplanned features. It is not unlikely that each basic model would have been prey to a different set of unintended social psychological disturbances.

In the clinical experiment a major source of such disturbances would have to be sought in the profound status differential that rigidly separated experimenters and subjects. This status differential meant that the experimental situation was, among other things, an encounter between individuals who were very unequal in power. Moreover, this power inequality was an *essential* feature of the experimental situation, because the subject's role was that of providing the material *on* which the experimenter could make his observations and test his hypotheses. The subject had to play the role of a biological organism or medical preparation if the experimental scenario was to be completed successfully. Finding oneself in this inferior position, the subject could be expected to be affected by the stresses and anxieties that people in such positions commonly experience. These subjects might have been expected to experience some anxiety about receiving a favorable evaluation from the experimenter and to worry about possible harmful consequences to themselves. A situation that they did not control and only half understood would have aroused anxiety in any case.

However, from the viewpoint of the outcome of the experiment, these anxieties were probably less serious in themselves than the various coping strategies in which the subjects, as persons in inferior and relatively powerless positions, could be expected to indulge.[23] These strategies might vary all the way from apathetic reactions to strenuous collusion with the investigator to give him the kind of performance he wanted. In any particular case one would not know how much of what was displayed to the investigator was due to peculiarities of specific coping styles and how much was due to the factors in which the investigator was really interested.

At this point we should not overlook the fact that the social structure of experimental situations may have had unintended social psychological consequences for the experimenter's as well as for the subject's role. People in positions of power and control are quite apt to drift into arrogance and callousness in the absence of appropriate institutional safeguards. Insofar as the social situation of the experiment is marked by a clear power differential, the psychological reactions of the more and the less powerful role incumbents are likely to be complementary to one another. So the performance the subject puts on for the benefit of the investigator is not recognized for what it is because the investigator does not think of the subject as a human agent but simply as material for the demonstration of his pet ideas. The extreme illustration of this outcome was provided by Charcot's notorious public demonstrations of hysteria at the Salpêtriere.[24] These hardly qualified as experiments, yet the social dynamics that they involved only represented gross forms of processes that had their subtler counterparts in every clinical experiment.

Although the investigative style pioneered by Galton would have shared some of the social psychological problems of the clinical experiment, the less-intense involvement of the participants and the introduction of a certain contractual reciprocity would tend to mitigate these problems. However, the kind of experimenter–subject relationship that this type of study promoted had potential problems of its own. In particular, the divergent interests that experimenters and subjects had in the outcome of the experimental situation, combined with the superficiality of their contact, contained the seeds of trouble. The subject's performance would be rendered in the service of aims that were not those of the investigator and the nature of their relationship was not such as to allow much opportunity for mutual explanations. This was surely a situation in which misunderstanding was likely to flourish. Such misunderstanding would take the form of divergent definitions of the experimental situation, and hence of the experimental task, by experimenter and subject. Although less significant for many of the tasks that Galton employed, it would become a more and more serious problem as Galton's successors increased the psychological complexity and potential ambiguity of experimental tasks.

Participants in the Leipzig model of psychological experimentation were much better protected against problems of this kind, for their collegial relationship and the exchangeability of experimenter and subject roles would ensure that misunderstanding was kept at a minimum. Nor would this style of research be plagued by the psychological consequences of extreme status inequality in the manner of the clinical experiment. However, the Leipzig model was not free of unwanted social psychological side effects of its own. The most noteworthy of these derived from the same source as its strength, namely, the close mutual understanding that developed among the participants. This made it relatively easy for observations to be reported that depended entirely on the conventions and tacit agreements among the members of specific research groups. Such observations were then unverifiable by anyone else. The reports from Titchener's Cornell laboratory on the nonexistence of imageless thought provided the classic example of this outcome.[25] It has been observed that the problems of "introspective psychology" did not arise on the level of comparing the introspective reports of individuals with each other but on the level of reconciling the conflicting claims of different laboratories. Historically, these problems had less to do with the intrinsic limitations of introspection than they had to do with the social psychological context in which introspection was employed – namely, the context provided by the Leipzig model of experimentation.

With the advantage of hindsight, it is easy to see that there was no prescription for arranging the social conditions of psychological experimentation that was free of unintended social psychological side effects. Psychological research was only able to collect its material by putting whole persons into social situations. Even though this research was originally interested neither in the persons nor in the social situation but in certain isolated intraindividual, processes, it could isolate these processes only conceptually. In practice, the research had to contend with the persons who were the carriers of these processes and with the social situations that were necessary to evoke the phenomena of interest. But that inevitably led to the appearance of various side effects, which were unintended and for a very long time not even recognized. However, the precise nature of these effects would vary with the model of investigative practice that was adopted. There was no single unflawed procedure for exposing the truth, only a variety of practices with different sources of error. Any estimate of the relative seriousness of these sources of error would have to depend on the kind of knowledge goal one had adopted.

The eclipse of the Leipzig model

The last three chapters have presented an analytic account of the origins of modern psychology as a form of investigative practice. In order to prepare the transition to the rest of this book, which will deal mainly with twentieth-century developments, let us take a brief look at certain historical developments beyond the earliest period of experimental psychology.

A major development was the rapid adoption of all the major techniques of psychological investigation by American investigators. The Leipzig model of investigative practice was replicated by many of those who had spent some time working with Wundt, most notably by Titchener at Cornell. But what made Titchener somewhat unusual was the fact that he never ceased to insist that the Leipzig model was the only one on which a genuine science of mental life could and should be based. Most of the early American investigators were much less dogmatic about their choice of methods, and some were extremely eclectic. G. S. Hall, for example, carried out different pioneering studies that followed all three of the models we have distinguished.[26] Cattell was particularly active in promoting the style of investigation that had been begun by Galton.[27] Clinical experimentation had its adherents too.[28] Early American psychology presented a mixture of investigative styles with few centers showing exclusive commitment to one style, although some had obvious preferences. However, in the course of the first half of the twentieth century a particular synthetic type of investigative practice developed and came to dominate American psychology. That development will be traced in the following chapters.

The model of investigative practice that proved least viable in the twentieth century was the Leipzig model. Because this model has a specific feature, which is always unambiguously reported in published research reports, it is possible to follow its decline rather accurately by means of a content analysis of psychological journals. The feature in question involves the exchange of experimenter and subject roles among at least some of the participants in the same experiment. Highly characteristic of the Leipzig model, the feature is clearly absent in clinical experimentation and in the type of investigation pioneered by Galton.

Concentrating on this feature as an index of the relative prevalence of the Leipzig model has the practical advantage that it is readily and reliably

Table 4.1. *Percentage of empirical studies in which there was an exchange of experimenter and subject roles*

Journal title	1894–1896	1909–1912	1924–1926	1934–1936
Philosophische Studien	32	79[a]	—[b]	—
American Journal of Psychology	38	28	20	8
Psychological Review	33	28	N.C.[c]	
Archiv für die gesamte Psychologie	—	43	28	16
Zeitschrift für angewandte Psychologie	—	4	N.C.	
Journal of Educational Psychology	—	2	0	0
Psychological Monographs	—	18	18	7
Journal of Experimental Psychology	—	—	4	6
Journal of Applied Psychology	—	—	0	0
Psychologische Forschung	—	—	38	31

[a]The second entry for *Philosophische Studien* is based on the new series of that journal, called *Psychologische Studien*.
[b]Dashes indicate periods before or after the appearance of a journal.
[c]An entry of N.C. means that coding of the journal was discontinued because no significant information was likely to result; for example, in the case of *Psychological Review* the proportion of codable empirical articles went from low to zero.

coded in published research reports. It should be noted, however, that this does not provide an accurate estimate of the *absolute* prevalence of the model, because it misses the cases where research colleagues exchange roles in different experiments but not in the same experiment. This practice was especially prevalent in the early days of experimental psychology, but picking it up from the research literature is time-consuming and hardly justified by the intrinsic interest of the results. There are also other features of the Leipzig model that are missed. However, it is the fluctuation in the *relative* popularity of the Leipzig model over the years that is historically more interesting, and this can be assessed by the simplified index as already described.

Empirical articles published in major American and German psychological journals were therefore coded in terms of the presence or absence of an exchange of experimenter and subject roles in the same study. (The sampling of journal volumes for different time periods, and the rules governing the inclusion of articles in the coding process are described in the Appendix.) Table 4.1 shows the results of this content analysis.

Although the simplified index used underestimates the use of the Leipzig model, this underestimation tends to be greatest when the use of this model is relatively high, especially in the earliest period. So the simplification in coding dilutes rather than accentuates the overall trend. This trend nevertheless emerges quite clearly. It involves an overall decline in the practice of exchanging experimenter and subject roles. Where continuity of publication allows a comparison of the same journal over several decades, the practice generally declines steadily with each time sample. The best documented case is that of the *American Journal of Psychology* where incidence of the practice gradually declines from 31 to 8 percent over a forty-year period. These figures do not suggest a sudden abandonment of the practice so much as a steady deterioration in its relative frequency of adoption. Nor does this general trend appear to be a specifically American phenomenon, because among the German journals, the one that is not associated with a particular school, *Archiv für die gesamte Psychologie,* shows the same trend.

It does seem as though the kind of experimental practice represented in this index was always somewhat more popular in German than in American psychology. In any time period the German figures are always higher than the American. By far the highest incidence (79 percent) of this pattern of research is to be found in *Psychologische Studien,* the Leipzig successor to Wundt's *Philosophische Studien.* But the Gestalt journal, *Psychologische Forschung,* also maintains a relatively high incidence on the index at a time when the practice of exchanging roles is being virtually abandoned by everyone else. At no time after 1890 did the original Leipzig style of arranging psychological experiments really dominate the discipline. At best it was one model among others. Never dominant, it had dwindled virtually to insignificance in American psychology during the period preceding World War II.

For applied psychology, whether American or German, it never had the slightest appeal, as is shown by the figures for the relevant journals, *Zeitschrift für angewandte Psychologie, Journal of Applied Psychology,* and *Journal of Educational Psychology.* Applied psychology had committed itself to knowledge goals that were unlikely to be advanced by the kind of investigative practice associated with Wundt's laboratory. What it was after was knowledge that could be quickly utilized by agencies of social control so as to make their work more efficient and more rationally defensible. Knowledge that led to behavioral prediction suited this purpose, but knowledge obtained in situations where the participants collaboratively explored the structure of their experience did not. Thorndike, a major pioneer of the new applied psychology, saw this very clearly when he complained that some of the procedures of academic experimental psychology "were not to aid the experimenter to know what the subject did, but to aid the subject to know what he experienced."[29] The kind of knowledge that Thorndike and others who shared his interests were after was predicated on a fundamental asymmetry between the subject and the object of knowledge. This asymmetry was not merely epistemic, however, but also social, because it was accepted that it was to be useful to the subject (or those for whom he acted) in controlling the object. What was desired was knowledge of individuals as *the objects of intervention rather than as the subjects of experience.* Experimental situations in which there was an exchange of roles between experimenters and subjects might well have seemed irrelevant to this type of knowledge goal.

As more and more academic psychologists, especially in the United States, adopted knowledge goals that were similar to those that had characterized applied psychology from the beginning, the Leipzig model was pushed even further into the background. It survived, on a relatively much reduced scale, only in the area of research on sensation and perception, the area in which it had also registered its first successes in the previous century.[30] Thus, as psychology's knowledge goals changed, its preferred mixture of investigative practices changed too.

Neither clinical experimentation nor the style of investigation pioneered by Galton was left unaffected by these changes. On the contrary, the Galtonian approach in particular underwent a dynamic development during the first half of the twentieth century. But this affected the nature of the knowledge produced rather than the social parameters of the experimenter–subject relationship. To analyze these developments, we will have to turn from the social interior of the investigative situation to its social exterior – that is, to the relations of investigators with each other and with the wider society. The making of a psychological knowledge product involved more than just the interaction of experimenters and subjects. It involved matters of conceptual form, of appropriate packaging, and of dissemination through approved channels. Twentieth-century developments in investigative practice mostly affected these aspects, and it is to them that we must now turn.

5

The triumph of the aggregate

Knowledge claims and attributions

Modern psychology may have started out with at least three different types of investigative practice, but, as we have seen, one of these hardly survived the first few decades in the history of the discipline. Among the other two, it was the Galtonian type that was to achieve a dominant position during the first half of the twentieth century. In the following chapters we will have to analyze various aspects of this complex historical development. Our analysis will draw heavily on the information contained in the psychological journals that now provide the major vehicle for the establishment of psychological knowledge claims. Twentieth-century psychology is a system of "public knowledge,"[1] and the major disciplinary journals constitute the crucial link in the transformation of the local knowledge produced in research settings into truly public knowledge.

This transformation involves the preparation of a research report that meets the criteria and conventions that prevail within the discipline or research area. Ostensibly, the research article communicates certain "findings" held to represent knowledge by the conventions of a particular community. In fact, the contents of the article only represent a knowledge *claim,* which may or may not be accepted by its intended audience. To convince this audience, the authors must submit to its conventions. For instance, they must adopt a certain literary style, relate the contents of the article to known concerns and interests of its possible audience, demonstrate the conformity of their methods to prevailing norms, and so on.[2] But the centerpiece of its work of persuasion is likely to involve the "findings" themselves, for they are after all the raison d'être of the whole paper. If the "results" that the author(s) wish to communicate do not measure up to prevailing standards of what constitutes significant knowledge in their